STA258: Statistics with Applied Probability
One Sample Hypothesis Tests: R Output

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1. A gourmet coffee roaster claims that their *single-origin* batch has an average caffeine content of 120 mg per cup. You sample 18 cups and measure the caffeine content (in mg) as follows:

data

110	118	95	122	130	107	115	109	113
108	124	100	116	114	105	121	98	111

Use the vector `coffee_data` in R. At the 5% level test whether the true mean caffeine content is less than 120 mg.

```
> library(mosaic)

> coffee_data <- c(110,118,95,122,130,107,115,109,113,108,124,100,116,
                   114,105,121,98,111)

> favstats(coffee_data)

   min      Q1 median     Q3    max    mean     sd  n missing
   95.0   105.8  111.0  118.5  130.0  112.00  10.24 18       0

> t.test(coffee_data, mu = 120, alternative = "less")

    One Sample t-test
    data: coffee_data
    t = -2.1692, df = 17, p-value = 0.0229
    alternative hypothesis: true mean is less than 120
    95 percent confidence interval:
       -Inf 117.19
    sample estimates:
    mean of x
       112.00
```

$\bar{x}$ s n

summary

σ unknown

outcome of a hyp-test

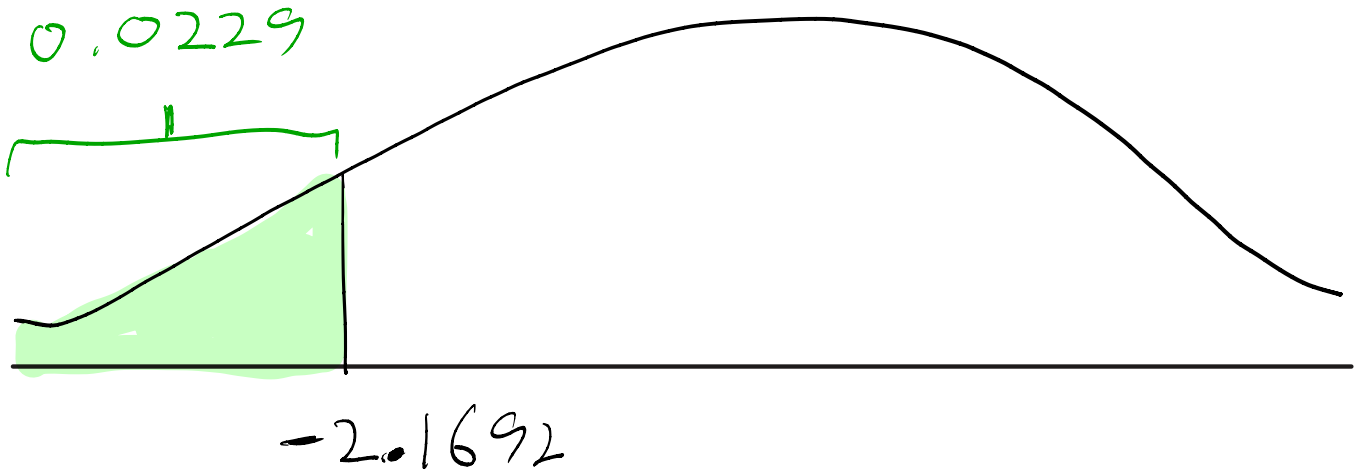
$H_0: \mu = 120$ $H_a: \mu < 120$

test stat: $t^* = -2.1692$

$t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{112 - 120}{10.24/\sqrt{18}}$

reference dist: t-dist at 17 df

p-value = 0.0229



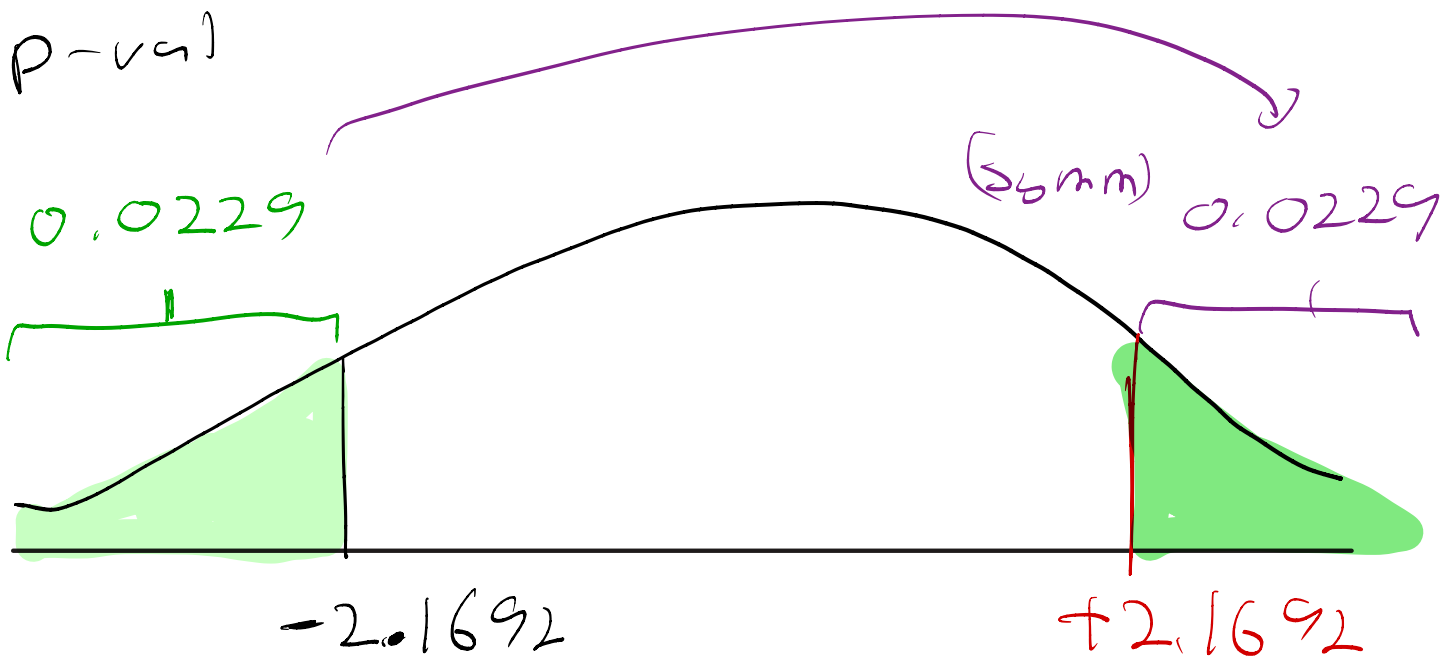
using output above, conduct the 2-sided test

$H_0: \mu = 120$ $H_a: \mu \neq 120$

test stat: $t^* = -2.1692$

reference dist: t-dist at 17 df

p-val



p-value $= 2 \times 0.0229$

$=$

2. A mobile-gaming company asserts that players spend on average 45 minutes per session in their flagship game. You record 20 sessions (in minutes) as follows:

52	40	49	60	45	51	38	47	56	43
50	46	57	44	53	48	41	55	47	49

Use the vector `game_sessions` in R. At the 1% level test whether the true mean session time is greater than 45 minutes.

```
> library(mosaic)

> game_sessions <- c(52, 40, 49, 60, 45, 51, 38, 47, 56, 43, 50, 46, 57,
                     44, 53, 48, 41, 55, 47, 49)

> favstats(game_sessions)

   min     Q1 median    Q3   max   mean     sd   n missing
   38.0  44.0   48.5  53.5  60.0  48.70   6.64  20       0

> t.test(game_sessions, mu = 45, alternative = "greater")
    One Sample t-test
    data: game_sessions
    t = 2.3854, df = 19, p-value = 0.0148
    alternative hypothesis: true mean is greater than 45
    95 percent confidence interval:
       46.80 Inf
    sample estimates:
    mean of x
       48.70
```

test stat

p-val

$H_a: \mu > 45$

3. In an urban-design study, a city-lab tests whether the average walk-to-bus-stop time in a new neighbourhood is 6 minutes. They sample 30 residents; times (in minutes) are:

6.4	4.8	5.9	5.0	4.7	6.8	5.1	4.9	5.3	5.6
5.0	4.2	6.1	5.4	4.3	5.8	4.6	5.2	5.0	4.9
6.0	5.5	4.4	5.1	5.9	4.8	5.3	4.7	5.6	5.4

Use the vector `walk_times` in R. At the 10% level test whether the true mean differs from 6 minutes.

```
> library(mosaic)

> walk_times <- c(6.4, 4.8, 5.9, 5.0, 4.7, 6.8, 5.1, 4.9, 5.3, 5.6, 5.0,
                  4.2, 6.1, 5.4, 4.3, 5.8, 4.6, 5.2, 5.0, 4.9, 6.0, 5.5,
                  4.4, 5.1, 5.9, 4.8, 5.3, 4.7, 5.6, 5.4)

> favstats(walk_times)

  min   Q1 median   Q3  max mean   sd  n missing
  4.2  4.8    5.2  5.6  6.8 5.20 0.99 30       0

> t.test(walk_times, mu = 6, alternative = "two.sided")

    One Sample t-test
    data: walk_times
    t = -4.7906, df = 29, p-value = 9.02e-05
    alternative hypothesis: true mean is not equal to 6
    95 percent confidence interval:
     4.84 5.56
    sample estimates:
    mean of x
      5.20
```

test stat

p-val

$H_a: \mu \neq 6$

$n-1 = 29$

$n = 30$

Recall $\chi^2_1 = Z^2$, $Z \sim N(0,1)$

$Z^* = \sqrt{2.3482} = \pm 1.5324$

4. A streaming platform claims that 65% of viewers finish an episode they start. You sample 170 viewers and find that 102 finished the episode. Use `finisher_count = 102`, `total_viewers = 170` in R. At the 5% level test whether the finishing proportion is less than 65%.

$\hat{p} = \frac{102}{170} = 0.60$

```
> prop.test(x = finisher_count, n = total_viewers,
            p = 0.65, alternative = "less", correct = FALSE)
```

$n = 170$

```
1-sample proportions test without continuity correction
data:  102 out of 170, null p = 0.65
X-squared = 2.3482, df = 1, p-value = 0.0620
alternative hypothesis: true p is less than 0.65
95 percent confidence interval:
  0.5487 0.6124
sample estimates:
   p
0.6000
```

test stat reported by R

$H_0: p = 0.65$ $H_a: p < 0.65$

5. A gourmet popcorn brand advertises "99% of users say it's perfectly popped." You survey 450 bags and find 442 are perfectly popped. Use `perfect_count = 442`, `bags_surveyed = 450` in R. At the 1% level test whether the true perfectly-popped proportion differs from 99%.

```
> prop.test(x = perfect_count, n = bags_surveyed,
          p = 0.99, alternative = "two.sided", correct = FALSE)

1-sample proportions test without continuity correction
data:  442 out of 450, null p = 0.99
X-squared = 7.8489, df = 1, p-value = 0.00508
alternative hypothesis: true p is not equal to 0.99
95 percent confidence interval:
  0.9570 0.9856
sample estimates:
   p
0.9822
```

6. A kombucha brewery states that the variance in alcohol-by-volume (ABV) across bottles from a single batch is exactly 0.04 (% ABV)^2. You sample 20 bottles and measure the following ABV values (in % ABV):

6.4	4.8	5.9	5.0	4.7	6.8	5.1	4.9	5.3	5.6
5.0	4.2	6.1	5.4	4.3	5.8	4.6	5.2	5.0	4.9
6.0	5.5	4.4	5.1	5.9	4.8	5.3	4.7	5.6	5.4

Use the vector `abv_values` in R. At the 2.5% significance level (two-sided) test whether the true variance differs from 0.04. Hint: use `library(EnvStats)` and `varTest()`.

```
> library(mosaic)
> library(EnvStats)

> abv_values <- c(0.45,0.50,0.43,0.48,0.46,0.52,0.49,0.47,0.51,0.44,
                  0.46,0.45,0.53,0.42,0.48,0.47,0.46,0.49,0.44,0.50)

> favstats(abv_values)
   min     Q1  median   Q3   max   mean     sd   n missing
  0.42   0.44    0.46 0.48  0.53  0.473  0.030  20     0

> varTest(abv_values, sigma.squared = 0.04, alternative = "two.sided")

$statistic
Chi-Squared
   0.434375

$parameters
df
19

$p.value
[1] 7.267107e-13

$estimate
    variance
0.0009144737

$null.value
variance
    0.04

$alternative
[1] "two.sided"
```

hyp test on variances

$\sigma_0^2 = 0.04$

$H_a: \neq$

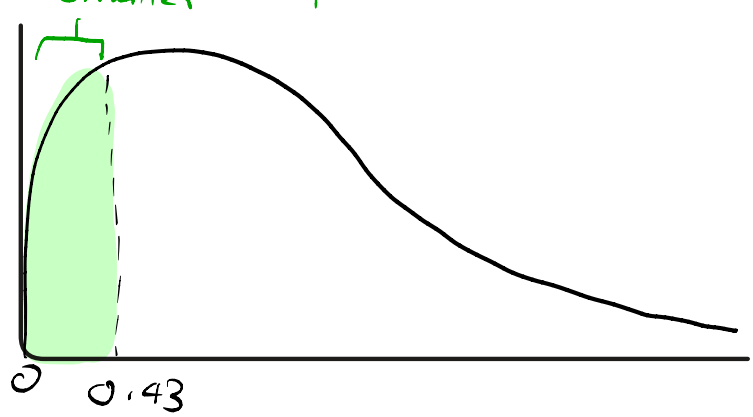
$H_a: \sigma^2 \neq 0.04$

2-sided

n-1 =

$\chi^{2*} = 0.434375$

smaller tail



p-val = 2 × smaller tail = 7.267×10^{-1}

```
$method
[1] "Chi-Squared Test on Variance"

$data.name
[1] "abv_values"

$conf.int
         LCL          UCL
0.0005288819 0.0019508188
attr(,"conf.level")
[1] 0.95

attr(,"class")
[1] "htestEnvStats"
```